

COVID-19 Trends, Vaccination & Mortality Analysis

Executive Summary

Objective

Analyze COVID-19 cases, deaths, vaccination rollout, and mortality patterns across countries and time using Power BI.

Key Findings

1. COVID-19 waves varied over time

Seven-day rolling averages reveal distinct periods of increased case activity, while smoothing daily reporting fluctuations makes major waves easier to identify.

2. COVID-19 impact differed across countries

Cases and deaths per million varied substantially between countries, showing why population-normalized measures are useful for cross-country comparisons.

3. Vaccination rollout progressed at different speeds

Countries reached vaccination coverage milestones at different times and levels, allowing the dashboard to compare rollout patterns over time.

4. Vaccination coverage and mortality show an observed relationship

The dashboard compares fully vaccinated coverage with deaths per million. The relationship is observational and does not establish that vaccination coverage alone caused differences in mortality.

5. Reported fatality and case growth changed over time

Reported case fatality rates and estimated doubling times varied across periods, reflecting changes in reported cases, deaths, and the timing of different waves.

Dashboard Analysis

The two-page Power BI dashboard provides:

COVID-19 Global Trends & Impact

Total cases and deaths

Cases and deaths per million

COVID-19 waves

Death trends

Country comparisons

Vaccination & Mortality Analysis

Vaccination rollout

Fully vaccinated coverage

Deaths per million

Reported case fatality rate

Case doubling time

Methodology

The dataset was cleaned using Power Query, with appropriate data types and missing-value handling. Seven-day rolling averages were used to reduce daily reporting fluctuations.

Population-normalized metrics were used for country comparisons, and vaccination coverage was calculated from vaccination counts and population.

Important Limitations

COVID-19 figures represent reported data, not necessarily all infections or deaths. Differences in testing, reporting practices, healthcare systems, population characteristics, and timing of waves can affect country comparisons. The vaccination analysis is observational and should not be interpreted as proof of causation.

Tools

Power BI | Power Query | DAX | Our World in Data

Overall: The dashboard provides an interactive, non-technical view of how COVID-19 waves, mortality, and vaccination rollout varied across countries and over time.